Workbook · Hand Calculation

Explainable Machine Learning Workbook

Six numerical examples you can follow without a calculator

Hand-calculation Workbook for XAI

Work through the main deck's (XAI_Lecture_Deck) formulas directly with small numbers.
Every example includes a verification step.

Six examples verify the main deck's key formulas by hand

Each example proceeds in three steps: setup, calculation, check

No.	Topic	Property checked	Main deck location
1	Shapley values of three features	Ordering weights, efficiency	PART 3
2	Linear SHAP	Closed form and additivity	PART 3
3	Permutation feature importance	Loss difference before and after shuffling	PART 2
4	Kernel SHAP weights	Large weights on small and large coalitions	PART 3
5	Counterfactual explanation	Finding the nearest alternative	PART 5
6	Integrated gradients	Completeness	PART 6

01

Example 1

Shapley Values of Three Features

Hypothetical default model: A delinquency history, B debt ratio, C income

Predicted default probabilities for eight coalitions are given

v(S) is the prediction knowing only feature set S. Knowing nothing gives the base value 0.10

Coalition S	v(S)	Coalition S	v(S)
⌀ (base value)	0.10	{A, B}	0.58
{A}	0.40	{A, C}	0.46
{B}	0.22	{B, C}	0.28
{C}	0.16	{A, B, C}	0.70

Goal: fairly divide the total increase 0.70 − 0.10 = 0.60 among A, B and C

With three features, the ordering weights are 1/3, 1/6, 1/3

|S| is the size of the coalition excluding i. There are two size-1 coalitions, each getting 1/6

$$w(|S|) = \frac{|S|!\,(3 - |S| - 1)!}{3!} \Rightarrow \; w(0) = \frac{2}{6} = \frac{1}{3}, \;\; w(1) = \frac{1}{6}, \;\; w(2) = \frac{2}{6} = \frac{1}{3}$$

Understanding via orderings

- Three features have 3! = 6 joining orders
- i comes first in 2 orders: 2/6
- i comes last in 2 orders as well: 2/6

Checking the weight sum

- 1/3 + 1/6 + 1/6 + 1/3 = 1
- The two size-1 coalitions share 1/6 each

The Shapley value of delinquency history A is 0.35

Weighted average of the increase from adding A to the four coalitions ø, {B}, {C}, {B,C}

$$\phi_A = \tfrac{1}{3}(0.40 - 0.10) + \tfrac{1}{6}(0.58 - 0.22) + \tfrac{1}{6}(0.46 - 0.16) + \tfrac{1}{3}(0.70 - 0.28)$$

$$= 0.10 + 0.06 + 0.05 + 0.14 = 0.35$$

Order	A's marginal contribution	Order	A's marginal contribution
A→B→C, A→C→B	0.40 − 0.10 = 0.30	C→A→B	0.46 − 0.16 = 0.30
B→A→C	0.58 − 0.22 = 0.36	B→C→A, C→B→A	0.70 − 0.28 = 0.42

Order check: (0.30 + 0.30 + 0.36 + 0.42 + 0.30 + 0.42) ÷ 6 = 2.10 ÷ 6 = 0.35

B is 0.17, C is 0.08, and the three values sum to 0.60

Compute B and C with the same weights, then check with the efficiency axiom

$$\phi_B = \tfrac{1}{3}(0.12) + \tfrac{1}{6}(0.18) + \tfrac{1}{6}(0.12) + \tfrac{1}{3}(0.24) = 0.04 + 0.03 + 0.02 + 0.08 = 0.17$$

$$\phi_C = \tfrac{1}{3}(0.06) + \tfrac{1}{6}(0.06) + \tfrac{1}{6}(0.06) + \tfrac{1}{3}(0.12) = 0.02 + 0.01 + 0.01 + 0.04 = 0.08$$

$$\phi_A + \phi_B + \phi_C = 0.35 + 0.17 + 0.08 = 0.60 = v(\{A, B, C\}) - v(\varnothing) = 0.70 - 0.10$$

Interpretation: delinquency history explains 58% (0.35) of the 0.60 increase. It is the #1 reason to notify the customer

02

Example 2

Linear SHAP

Compute directly as coefficient × (value − mean)

In a linear model, SHAP values take one line

The base value φ_0 is the prediction at the mean input

$$\hat{f}(x) = 0.5 + 2x_1 - 1\,x_2, \qquad E[x_1] = 1,\ E[x_2] = 3, \qquad x = (2,\ 1)$$

$$\phi_1 = 2\,(2-1) = +2, \qquad \phi_2 = -1\,(1-3) = +2, \qquad \phi_0 = \hat{f}(E[x]) = 0.5 + 2 - 3 = -0.5$$

$$\phi_0 + \phi_1 + \phi_2 = -0.5 + 2 + 2 = 3.5 = \hat{f}(x) = 0.5 + 4 - 1$$

Why is x_2 positive?

- The coefficient is negative, but the value is below the mean, so it raised the prediction

Lesson

- Don't judge the direction of contribution by the coefficient's sign alone

03

Example 3

Permutation feature importance

Compare the loss before and after shuffling with four observations

Shuffling a used feature increases loss; an unused feature scores 0

The model $f(x) = 2x_1$ does not use x_2. The loss is mean squared error (MSE)

Obs.	x_1	x_2	y	Original prediction	Shuffled x_1	Prediction after shuffling	Error
1	1	5	2	2	2	4	+2
2	2	1	4	4	1	2	−2
3	3	4	6	6	4	8	+2
4	4	2	8	8	3	6	−2

$$FI_1 = \mathrm{MSE}_{\text{shuffled}} - \mathrm{MSE}_{\text{original}} = \frac{2^2 + 2^2 + 2^2 + 2^2}{4} - 0 = 4, \qquad FI_2 = 0 - 0 = 0$$

Shuffling x_2 leaves predictions unchanged, so its importance is 0

04

Example 4

Kernel SHAP weights

Weights by coalition size with four features

Small and large coalitions weigh twice as much as middle ones

Plug M = 4 into π(z′) = (M−1) / [C(M,|z′|)·|z′|·(M−|z′|)]

$$M = 4: \quad \pi(1) = \frac{3}{4 \cdot 1 \cdot 3} = 0.25, \qquad \pi(2) = \frac{3}{6 \cdot 2 \cdot 2} = 0.125, \qquad \pi(3) = \frac{3}{4 \cdot 3 \cdot 1} = 0.25$$

Why

- Coalitions with just one feature on or just one off show that feature's effect most cleanly
- Size-0 and size-4 coalitions have infinite weight and are handled as constraints

Practical implications

- With a small sampling budget, draw size-1 and size-3 coalitions first
- shap's KernelExplainer actually selects samples this way

05

Example 5

Counterfactual explanation

Find the nearest approval condition for a declined customer

Repaying 10 million won of debt is the nearer counterfactual

Hypothetical score model. Income and debt in units of 10 million won. Customer with income 5, debt 1.5

$$z = -2 + 0.5 \times \text{income} - 1.0 \times \text{debt}, \qquad \text{approve if } z \geq 0$$

$$z = -2 + 0.5(5) - 1.0(1.5) = -1.0 \Rightarrow \text{decline}$$

Alternative	Value changed	Required change	Size of change
Reduce debt only	−1.0 × debt = −0.5	1.5 → 0.5 (repay 10 million won)	1.0
Raise income only	0.5 × income = 3.5	5 → 7 (20 million won increase)	2.0

If distances share units, repaying debt (the smaller change) is the nearer counterfactual. In practice, standardize units

06

Example 6

Integrated gradients

Verify completeness in a multiplicative model

The integrated gradients sum exactly to the change in prediction

On the path $x' + \alpha(x - x') = (2\alpha, 3\alpha)$, $\partial f/\partial x_1 = x_2 = 3\alpha$ and $\partial f/\partial x_2 = x_1 = 2\alpha$

$$\hat{f}(x) = x_1 x_2, \qquad x = (2, 3), \qquad x' = (0, 0)$$

$$IG_1 = 2 \int_0^1 3\alpha \, d\alpha = 2 \times 1.5 = 3, \qquad IG_2 = 3 \int_0^1 2\alpha \, d\alpha = 3 \times 1 = 3$$

$$IG_1 + IG_2 = 6 = \hat{f}(2, 3) - \hat{f}(0, 0)$$

Since the two features multiply symmetrically, 6 splits equally into 3 and 3. The Shapley values are also 3 and 3

Three properties confirmed by the six examples

A recap before returning to the main deck

1

It adds up

Shapley values, linear SHAP and integrated gradients all split the change in prediction completely.

2

Sign depends on value versus mean

Even with a negative coefficient, a value below the mean raises the prediction.

3

Explanations lead to action

Counterfactual explanations show the customer the smallest change that flips the outcome.